Angles of Reflection

Aim: To observe the relationship between angles of incidence and reflection.

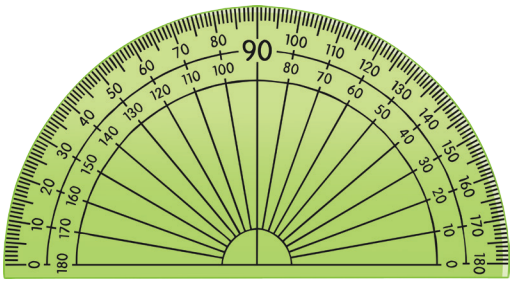
Equipment:

- ray box
- slit plate
- mirror
- lab pack (if needed)
- protractor
- sharp pencil
- ruler
- activity sheet

Safety Information

Filament bulb – risk of burns

Handle with extra care. Rinse any burns under cold running water for 10 minutes and inform the teacher immediately.



Method:

Step 1: On the activity sheet, set up the equipment as shown in the diagram.

Step 2: Place the mirror onto the indicated space on the sheet.

Step 3: Direct the ray of light along the angle indicated by the letter A.

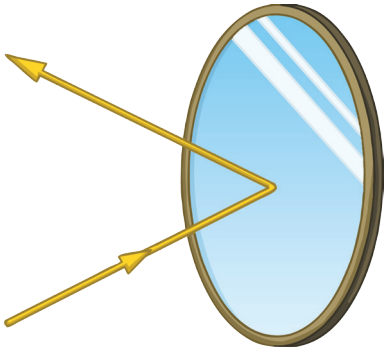
Step 4: Calculate the angle of incidence.

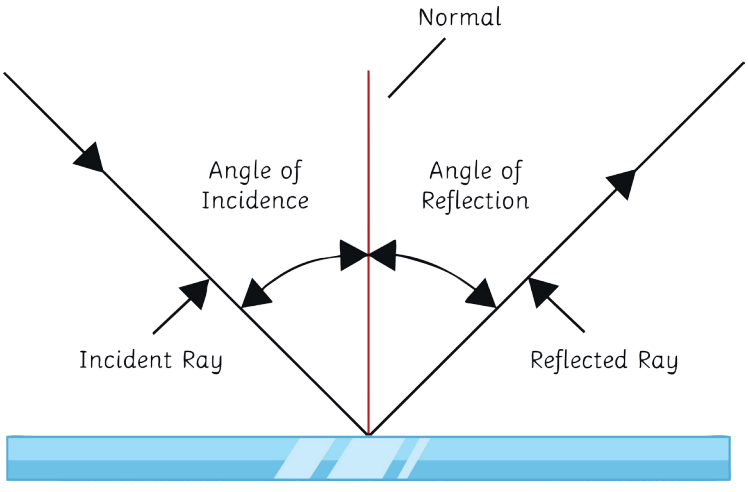
Step 5: On the other side of the sheet, draw a line where the ray of light emerges and label this with the letter A.

Step 6: Repeat steps 3 and 4 for each of the other angles B to H.

Step 7: Remove the mirror and position the protractor over the activity sheet.

Step 8: Measure the angle between the normal and each of the reflection angles you plotted and labelled A to H in steps 3 to 5. Record your findings in the table.





Results

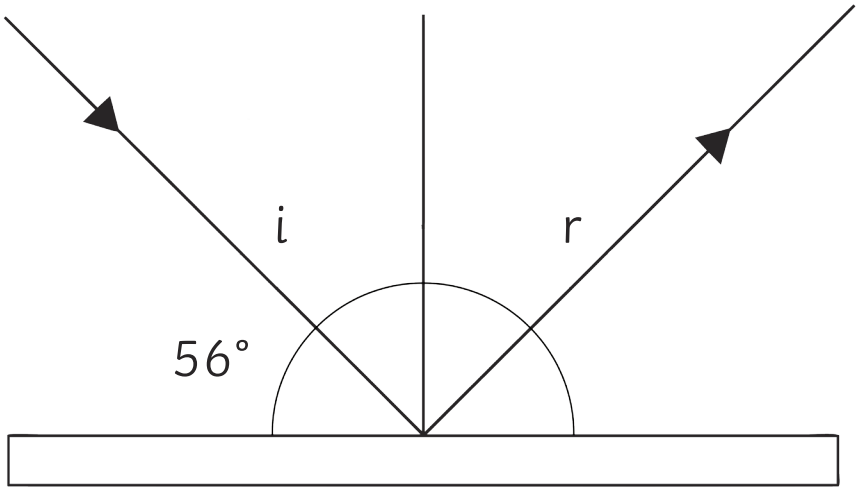
	Angle of Incidence	Angle of Reflection
A	80˚	
B	70˚	
C	60˚	
D	50˚	
E	40˚	
F	30˚	
G	20˚	
H	10˚	

Questions:

1. Describe the relationship between the angle of incidence and the angle of reflection.

2. If the incidence ray was along the normal (perpendicular to the mirror), what would the angle of reflection be?

3. Look at the ray diagram below. What is the angle of reflection? (Diagram not to scale).



Results **Answers**

	Angle of Incidence	Angle of Reflection
A	80˚	80˚
B	70˚	70˚
C	60˚	60˚
D	50˚	50˚
E	40˚	40˚
F	30˚	30˚
G	20˚	20˚
H	10˚	10˚

Questions:

1. Describe the relationship between the angle of incidence and the angle of reflection.

The angle of reflection is equal to the angle of incidence.

2. If the incidence ray was along the normal (perpendicular to the mirror), what would the angle of reflection be?

90˚, it would reflect directly back along the same path of the ray of incidence.

3. Look at the ray diagram below. What is the angle of reflection? (Diagram not to scale).

34˚

Activity Sheet

